

- 1 (a) The golden mantella, *Mantella aurantiaca*, is a small terrestrial frog found in Madagascar.

Fig. 1.1 shows a golden mantella.



Fig. 1.1

- (i) Name the domain and kingdom to which the golden mantella belongs.

domain

kingdom

[2]

- (ii) The skin of the golden mantella is brightly coloured and contains a toxin.

Suggest a benefit to the frog of being brightly coloured.

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..... [1]

- (b) The toxin in the skin of the golden mantella affects the action of the sarcomeres in muscle fibres (muscle cells) of mammalian striated muscle. The toxin inhibits a protein, $\text{Ca}^{2+}\text{ATPase}$, found in the membrane of the sarcoplasmic reticulum.

$\text{Ca}^{2+}\text{ATPase}$ pumps calcium ions from the cytoplasm into the sarcoplasmic reticulum when the fibre is no longer stimulated.

Suggest the consequences to the sarcomere of the action of the golden mantella toxin.

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(c) Describe the role of calcium ions in a cholinergic synapse.

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[Total: 9]